

Technical Brief: Semantic Communication Frameworks for Task-Oriented Industrial Surface Anomaly Detection

Focus on PCB Defect Localization (Open, Short, Mousebite)

Project Group Technical Report

Course: Final Year Capstone Project

Target Platform: Overleaf & LaTeX Compilation

May 27, 2026

Abstract

This technical report establishes the foundational paradigm shift from classical Shannon information theory to task-oriented Semantic Communication (SemCom) within automated industrial visual inspection systems. Specifically optimized for Printed Circuit Board (PCB) surface defect detection—categorized into micro-scale geometric anomalies including open circuits, short circuits, and mousebites—this document provides a rigorous architectural analysis. We formalize five semantic communication archetypes, map their mathematical frameworks, evaluate trade-offs in channel bandwidth, and propose an optimal end-to-end framework integrating Deep Joint Source-Channel Coding (Deep JSCC) with multi-scale feature fusion for deployment over band-limited factory wireless networks.

1 Introduction & Paradigm Shift

Traditional industrial visual inspection pipelines rely heavily on high-resolution automated optical inspection (AOI) systems. In a classic centralized architecture, edge-mounted smart cameras capture high-resolution imagery of printed circuit boards and transmit raw or source-compressed data (e.g., JPEG, PNG) to a central computing cluster for defect inference. This methodology is bounded by **Shannon's Classical Information Theory**, which characterizes communication purely as the accurate reconstruction of a sequence of bits across a noisy channel, governed by the Shannon capacity formula:

$$C = B \log_2 \left(1 + \frac{P}{N_0 B} \right) \quad (1)$$

where C is the channel capacity, B is the bandwidth, P is the signal power, and N_0 is the noise power spectral density.

However, in an ultra-dense Industrial Internet of Things (IIoT) network operating multiple parallel inspection lines, transmitting raw pixel data strains physical network limits, causing high latency, packet drops, and heavy infrastructure costs. Furthermore, pixel-level accuracy is redundant when the downstream goal is strictly diagnostic inference.

Semantic Communication (SemCom) moves beyond the bit-level accurate reconstruction paradigm to the *semantic level*, where the primary goal is the successful transmission of the meaning, context, or utility of the data relative to a specific task. By executing task-oriented feature extraction directly at the edge camera, background noise, invariant substrate pixels, and uniform copper regions are dropped, transmitting only the critical feature vectors required to detect and localize structural anomalies.

2 Mathematical Formulation of Task-Oriented SemCom

To formalize a task-oriented semantic communication system for PCB defect classification, let the raw input image captured by the edge camera be represented by the random variable $X \in \mathcal{X}$. The downstream task label (e.g., normal, open, short, mousebite) is defined as $Y \in \mathcal{Y}$.

The transmitter uses a deep neural network parameterized by θ to extract a highly compressed semantic representation vector $Z = f_\theta(X)$. This vector is transmitted across a noisy physical channel, yielding the received feature map $\hat{Z}$ at the server. The receiver decodes this to perform task classification $\hat{Y} = g_\phi(\hat{Z})$, where ϕ represents the decoder weights.

The optimization space is governed by the **Information Bottleneck (IB) Principle**, which aims to minimize the information transmitted across the channel while maximizing the mutual information relative to the downstream task Y :

$$\min_{\theta, \phi} \mathcal{L}_{IB} = I(X; Z) - \beta I(Y; \hat{Z}) \quad (2)$$

where $I(\cdot; \cdot)$ denotes mutual information, and β is a Lagrange multiplier controlling the trade-off between the compression rate and task accuracy. The semantic loss profile shifts from Mean Squared Error (MSE) on pixels to cross-entropy loss on task labels:

$$\mathcal{L}_{\text{Task}} = - \sum_{i=1}^{|\mathcal{Y}|} Y_i \log \hat{Y}_i \quad (3)$$

3 Five SemCom Archetypes for Industrial Visual Quality Control

The implementation of semantic communications for PCB inspection can be mapped into five distinct architectural frameworks based on operational requirements, network constraints, and human-in-the-loop dependencies.

3.1 Pure Task-Oriented Semantic Communication (Non-Reconstructive)

In this paradigm, the physical reconstruction of the PCB image is completely abandoned. The system focuses entirely on optimizing downstream semantic inference.

- **Operational Workflow:** The edge device extracts multi-scale deep features from the PCB image using a deep convolutional or transformer-based backbone, maps these features directly into a compact, low-dimensional semantic vector Z , and transmits them. The backend server maps Z directly to an object detection or segmentation head.
- **Data Structure and Footprint:** The transmitted payload consists of a continuous floating-point matrix representing non-redundant semantic topologies. Bandwidth consumption drops by up to 99.5% compared to raw image transmission.
- **Defect Mapping Context:** Ideal for high-speed automated sorting arms where real-time latency is paramount. The system outputs an instantaneous text array: {"Defect": "Mousebite", "Coor

3.2 Semantic Reconstruction-Oriented Communication

When human engineers must inspect flagged anomalies on a central operations dashboard to visually verify structural integrity before discarding expensive multilayers, generative pixel reconstruction is required.

- **Operational Workflow:** The semantic vector is optimized using a perceptual loss function alongside a task loss. The server utilizes a generative decoder framework—such as Generative Adversarial Networks (GANs) or conditional Latent Diffusion Models (LDMs)—to synthesize the image back from the semantic latent code.

- **Data Structure and Footprint:** Slightly higher footprint than pure task-oriented systems, as it must encapsulate structural geometry parameters alongside diagnostic task indicators.
- **Defect Mapping Context:** The generated image minimizes semantic distortion over pixel distortion. The green substrate or minor background variations may appear smoothed or generic, but the critical boundaries—such as the exact missing copper profile of a mousebite or the bridging metallic filaments of a short circuit—are rendered with crisp edge-fidelity.

3.3 Region of Interest (ROI) / Segmentation-Driven SemCom

PCB surface structures are sparse; a significant portion of the image area consists of non-functional substrate or un-routed isolation zones. This paradigm leverages spatial semantic awareness to selectively allocate bandwidth.

- **Operational Workflow:** A lightweight, highly optimized semantic segmentation head runs locally on the edge camera to generate an immediate binary or multi-class semantic mask separating Regions of Interest (ROI: copper traces, annular rings, solder pads) from Regions of Non-Interest (RONI: bare FR-4 substrate).
- **Data Structure and Footprint:** Dynamic bandwidth allocation. High-frequency structural components are transmitted at maximum bit-depth, while the background matrix is heavily compressed or discarded entirely.
- **Defect Mapping Context:** Ensures that fine geometric boundaries remain uncompromised. Since a mousebite represents a tiny indentation on a trace border, maintaining edge fidelity within the trace ROI prevents false negatives.

3.4 Deep Joint Source-Channel Coding (Deep JSCC)

Traditional systems split communication into source coding (compression via JPEG/H.264) and channel coding (error-correction via Turbo/LDPC codes). Deep JSCC breaks this boundary by mapping source features directly into physical channel symbols using end-to-end neural networks.

- **Operational Workflow:** The Deep-Vision Fusion feature extractor processes the PCB image. The resulting feature map is mapped directly into continuous complex channel input symbols $X_c \in \mathbb{C}^K$ by an analog-like encoder layer. These symbols are directly transmitted over the air.
- **Data Structure and Footprint:** The payload is represented as raw baseband I/Q modulation symbols matching the channel’s physical degrees of freedom.
- **Defect Mapping Context:** Industrial environments suffer from severe electromagnetic interference caused by high-power motors and machinery. While traditional digital communication suffers from a abrupt connection loss (cliff effect) when signal quality drops below a threshold, Deep JSCC exhibits graceful degradation. Even at ultra-low Signal-to-Noise Ratios (SNR), the semantic meaning of a catastrophic short circuit passes through safely.

3.5 Knowledge-Base (KB) Driven Semantic Communication

Industrial manufacturing is highly structured: every PCB batch follows an exact, deterministic Computer-Aided Design (CAD) layout or a verified "Golden Template" design matrix.

- **Operational Workflow:** Both the edge camera (transmitter) and the central server (receiver) store a synchronized local copy of the flawless CAD reference layout. The transmitter aligns the test image to the template and calculates a spatial structural residual matrix.

- **Data Structure and Footprint:** Under normal conditions, the transmitted payload approaches zero bits. Only semantic anomalies (deviations from the KB template) trigger transmission.
- **Defect Mapping Context:** When a trace breaks ("open circuit"), the edge device detects a localized geometric divergence from the knowledge base, encoding and transmitting only the bounding parameters and morphology of that specific break.

4 Proposed System Architecture for the Capstone Project

To optimize a final year project that combines computer vision with 6G SemCom concepts, a hybrid architecture incorporating **Pure Task-Oriented Feature Extraction with Deep JSCC** is highly recommended.

Table 1: Comparative Matrix of Semantic Communication Paradigms for PCB Inspection

Paradigm Type	Bandwidth Usage	Computational Overhead	Reconstruction	Primary Failure
Pure Task-Oriented	Ultra-Low ($< 1\%$)	Medium (Edge Encoder)	None	Semantic Drift
Semantic Reconstruct.	Low ($5 - 15\%$)	High (Server LDM/GAN)	Generative	Hallucinated Defects
ROI / Seg-Driven	Variable ($10 - 40\%$)	Low (Binary Masking)	Hybrid Pixel	Mask Misalignment
Deep JSCC	Static Channels	Medium (End-to-End)	None	Deep Fading Dropouts
Knowledge-Base Driven	Minimal ($< 0.1\%$)	High (Spatial Alignment)	Residual Only	Template Calibration

4.1 The Transmitter: Semantic-Aware Deep-Vision Fusion Encoder

The encoder utilizes a Feature Pyramid Network (FPN) coupled with a Spatial-Channel Attention block (e.g., CBAM module). This structure extracts both local high-frequency geometric boundaries (critical for spotting mousebites) and global semantic trace continuity (critical for spotting opens and shorts). The output feature tensor is compressed into a compact bottleneck representation.

4.2 The Physical Simulation Layer: Noisy Communication Channel

To demonstrate the communication aspect in standard deep learning environments (e.g., PyTorch), the transmitted vector Z is subjected to an explicit channel degradation block simulating an additive white Gaussian noise (AWGN) channel model:

$$\hat{Z} = HZ + N, \quad N \sim \mathcal{N}(0, \sigma^2) \quad (4)$$

By explicitly altering the noise variance σ^2 during training, the model learns to extract deep structural features that are highly robust to physical transmission interference.

4.3 The Receiver: Task-Oriented Downstream Detector

The corrupted received feature map $\hat{Z}$ is fed directly into a multi-head network layout consisting of a classification branch (anomaly categorization) and an optional bounding box regression branch for defect coordinate localization.

5 Conclusion

Transitioning to a Task-Oriented Semantic Communication framework enables next-generation industrial inspection networks to achieve low latency and robust anomaly detection. By implementing a joint source-channel semantic architecture, this project successfully bridges advanced deep learning feature fusion with the emerging paradigms of 6G communication networks.

References

- [1] F. Liu, W. Tong, Y. Yang, Z. Sun, and C. Guo, “Task-oriented image semantic communication based on rate-distortion theory,” *arXiv preprint arXiv:2201.10929*, 2022.
- [2] S. Yao, Y. Lu, K. Niu, J. Dai, C. Dong, and P. Zhang, “Semantic information processing for interoperability in the Industrial Internet of Things,” *Fundamental Research*, vol. 4, no. 1, pp. 8–12, 2024.
- [3] C. Guo, J. Xi, Z. He, J. Liu, and J. Yang, “A knowledge base driven task-oriented image semantic communication scheme,” *Remote Sensing*, vol. 16, no. 21, p. 4044, 2024.